# PNCC Care

## Engineering Dignity: Senior Safety, Daily Check-In & Caregiver Companion

A Complete Technical Dossier, Comprehensive Architectural Analysis, and Fully Illustrated Non-Technical User Manual.
Features complete breakdowns of every setting, functional consequences, failure modes, clinical caregiver protocols, and multi-device IoT integration.

📱 iPhone, iPad & Android
⌚ watchOS Standalone Companion
🗣️ Amazon Alexa Voice Check-In
🌙 Bedside Nightstand Ambient Clock
🛡️ 2-Tier Timers & Fall Sensor
🔒 100% Private / Zero Ad Trackers

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**Product Version:** v1.1.5 (Build 25) · **Release Date:** November 2026
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**App Store ID:** 6742517865 · **App ID:** com.zigzagyc.dailyNotify

CHAPTER 1 · DESIGN OBJECTIVE & PHILOSOPHY

# The Target of the Design: Reconciling Family Care with Elder Dignity

> Resolving the heartbreaking modern dilemma: adult children burdened by constant fear for solitary parents, versus proud elders who refuse invasive surveillance and complicated gadgets.

## 1.1 The Underlying Societal Crisis

Throughout the world, rapid urbanization and longer lifespans have created millions of "solitary elders" living independently. While independence is cherished, it creates a persistent, silent panic among adult children:

- *"Did Mom get out of bed this morning?"*
- *"What if Dad slipped in the shower last night and cannot reach his phone?"*
- *"If an accident occurred, how many hours or days would pass before anyone noticed?"*

## 1.2 Why Conventional Monitoring Solutions Fail

Traditional attempts to solve this problem create heavy emotional and practical friction:

- **Surveillance Cameras:** Installing cameras in the living room or bedroom strips the elder of privacy and self-respect. Elders feel imprisoned, watched, and patronized.
- **Daily Interrogation Calls:** Calling daily just to ask "Are you alive? Did you take medicine?" turns loving relationships into clinical inspections, triggering irritation and resistance.
- **Medical Panic Buttons / Pendants:** Elders frequently leave emergency necklaces in drawers because wearing them in public feels like wearing a badge of disability.

## 1.3 The Design North Star: "Passive-Active Balance"

**PNCC Care was founded on a simple decree: A safety check-in should not feel like an inspection; it must be an uplifting, effortless morning ritual taking less than 2 seconds.**

Instead of requiring the senior to master complex technology, PNCC Care aligns with the natural daily rhythms of life: waking up, looking at an ambient bedside clock, wearing an Apple Watch, reading a soothing Bible scripture, or speaking casually to a smart speaker from bed.

CHAPTER 2 · ACCOUNT, LOGIN & ROLE ARCHITECTURE

# Account Registration, Profile Selection & Switching

> Password-free biometric onboarding and flexible multi-generation profile switching.

## 2.1 Password-Free Biometric Authentication

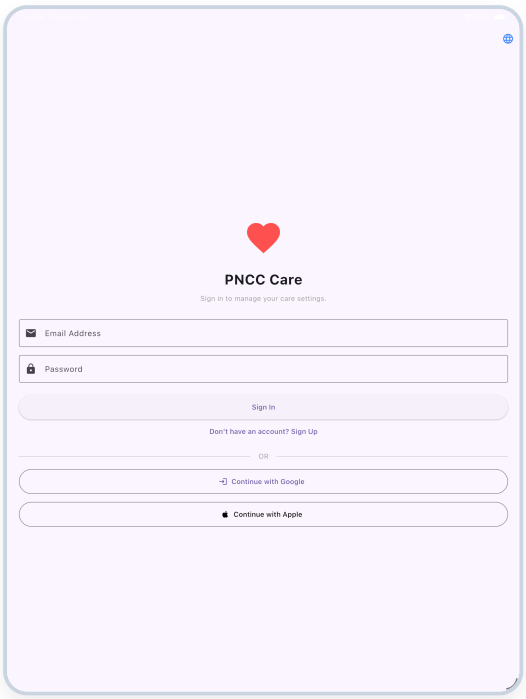



Figure 2.1: Multi-Platform Biometric Login (Apple, Google & Family Email)

To eliminate the number-one reason seniors abandon apps (forgotten passwords), PNCC Care provides three direct sign-in paths:

- **Sign in with Apple (Recommended for iOS/iPadOS):** One tap authenticates via Face ID or Touch ID. Generates an encrypted, private Apple relay token. Zero passwords to memorize.
- **Sign in with Google (Recommended for Android):** Direct OAuth2 linking to existing Android Google accounts with single-tap authentication.
- **Email & Password:** Standard email registration (minimum 6-character password) for families who share a dedicated non-OAuth family email account.

## 2.2 The Three User Roles: Senior, Caregiver & Dual Mode




Figure 2.2: Profile Selection (Senior, Caregiver & Dual Mode)

| User Role | Target User & Ergonomics | Screen Layout & Functional Consequence |
|---|---|---|
| **Senior Mode (長輩模式)** | For aging parents. Maximum accessibility, minimum cognitive load. | Displays the giant green **"I'm Okay"** card, real-time safety status indicator, daily scripture, and bedside nightstand clock. Suppresses all administrative menus. |
| **Caregiver Mode (照護者模式)** | For adult children, family members, nurses, and church volunteers. | Displays the comprehensive Caregiver Console. Shows real-time color-coded cards for multiple elders, countdown timers, last check-in timestamps, and emergency phone/SMS shortcuts. |
| **Dual Mode (雙向模式)** | For the "Sandwich Generation" (e.g. 60-year-old checking in for their adult kids while caring for their 85-year-old mother). | Adds a bottom navigation bar. Tap **Senior** to check in for yourself; tap **Caregiver** to monitor your aging parents. |

**How to Switch Roles Anytime:** Tap **Settings (⚙️)** in the top-right corner → scroll down to **"Switch Profile" (⇄)** → tap your desired card. The entire app layout transforms instantly without signing out!

CHAPTER 3 · THE SENIOR EXPERIENCE

# The Senior Dashboard & Six Multimodal Check-In Triggers

> Six distinct ergonomic pathways to check in so the senior never has to "learn" a mobile app.

## 3.1 Senior Dashboard Anatomy

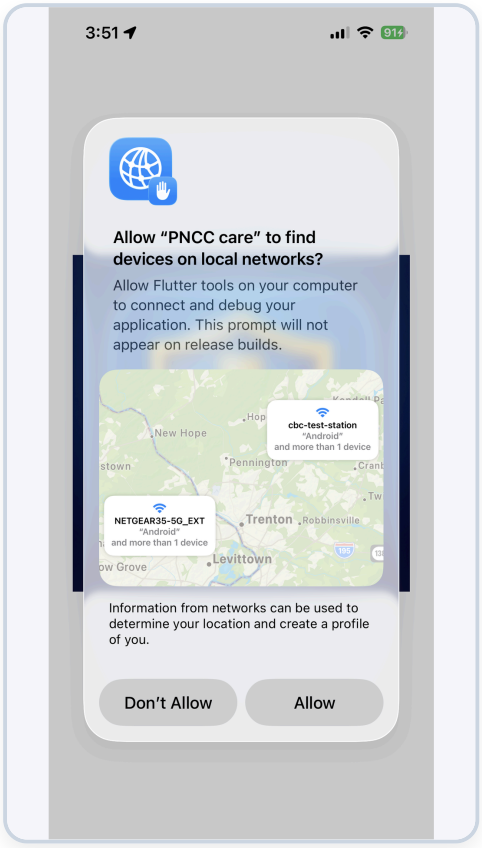



Figure 3.1: Senior Dashboard with Giant Card & Safe Status Indicator

The Senior Dashboard is intentionally constrained to a single, non-scrolling primary viewport:

- **Consolidated Safety Card:** Merges status ("All is Well") and touch target into one prominent 48dp+ high-contrast card.
- **500ms Auto-Dismissal:** Tapping the button triggers a physical haptic vibration, displays an animated checkmark, and minimizes/closes the app automatically after 500 milliseconds. The elder never feels stranded in the app.

## 3.2 Detailed Breakdown of All Six Check-In Triggers

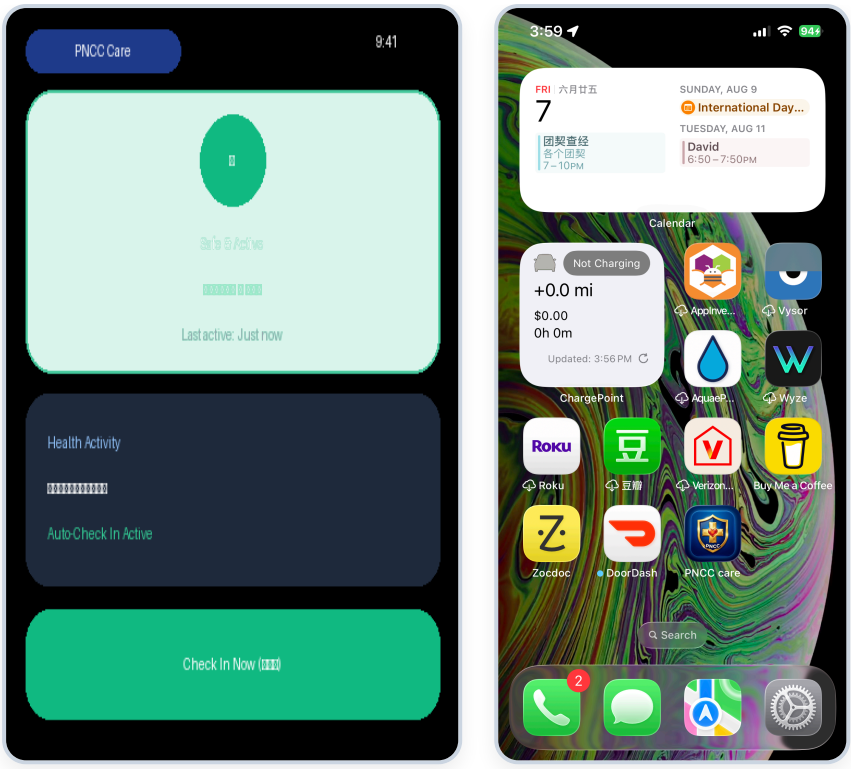



Figure 3.2: Apple Watch Wrist Companion (Left) & Home Screen Widgets (Right)

| Trigger Modality | How to Use | Functional Consequence & Architectural Behavior |
|---|---|---|
| **1. Smartphone Giant Button** | Open app, tap the green card. | Resets Timer 1 and Timer 2, writes timestamp to Cloud Firestore, updates home screen widget, triggers haptics, and auto-exits. |
| **2. Apple Watch Companion** | Glance at wrist, tap "I'm OK · 我很好". | Sends encrypted packet via WatchConnectivity (WCSession) to iPhone. If iPhone is out of range, caches locally and syncs upon reconnection. |
| **3. Amazon Alexa Voice Skill** | Speak aloud: *"Alexa, tell Daily Notify I'm OK."* | Echo smart speaker contacts AWS Lambda backend, authenticates 6-digit device token, resets cloud timer, and Alexa recites today's Bible verse in English or natural Mandarin Chinese (AWS Polly Zhiyu). |
| **4. Bedside Nightstand Screen Saver** | Phone placed in charging dock; touch the dark clock screen. | Displays ambient OLED clock and scripture. Touching anywhere dismisses the clock AND silently registers today's check-in! |
| **5. Desktop & Lock Screen Widgets** | Tap green widget button on home screen without opening the app. | iOS WidgetKit / Android AppWidgetProvider intercepts background tap, updates local storage, syncs to cloud, and flips widget to "Safe". |
| **6. Sensor Fall Detection** | Passive. Phone senses heavy impact. | Triggers local alarm siren and displays 30-second red cancel countdown. If not dismissed within 30 seconds, dispatches emergency alert. |

CHAPTER 4 · COMPLETE SETTINGS & CONSEQUENCES

# Exhaustive Settings Reference: Every Option, Value & Functional Consequence

> A comprehensive technical manual of every toggle, slider, timer, and option in PNCC Care Settings, detailing the exact system consequences of each choice.

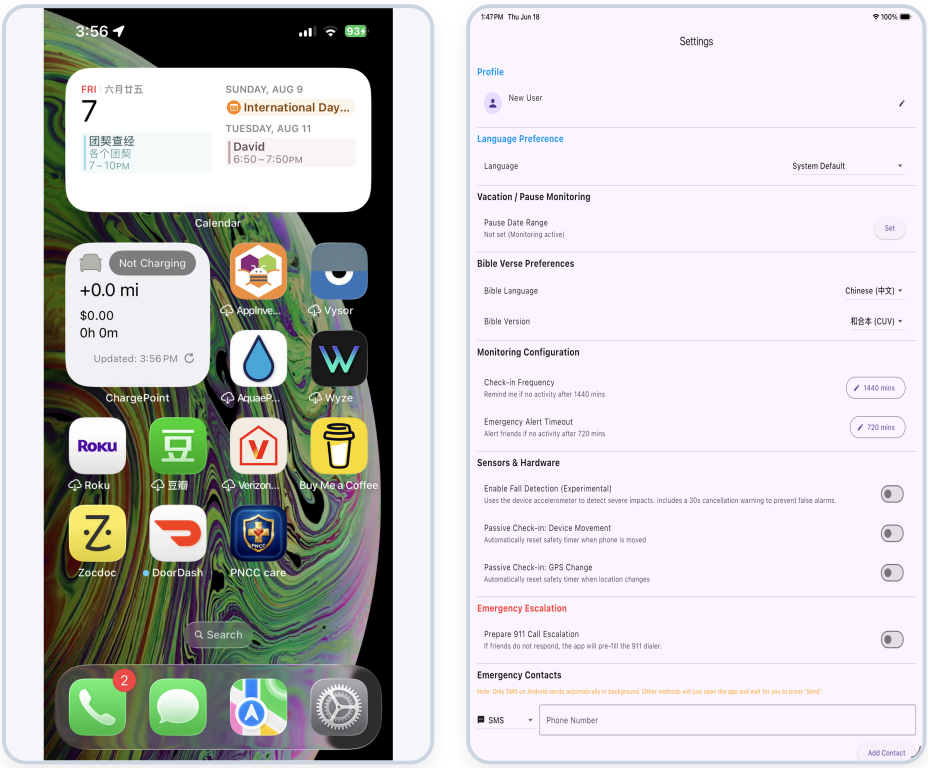



Figure 4.1: Settings Navigation on Phone (Left) and Detailed iPad View (Right)

## 4.1 Timer & Inactivity Configurations

| Setting Name | Options & Ranges | Detailed Mechanism & Functional Consequence |
|---|---|---|
| **Check-In Frequency (Timer 1)** | 60 to 2880 mins (Default: 1440m / 24h) | **Mechanism:** The active "Safe" window countdown. **Consequence:** When Timer 1 expires, the status card changes from Green (Safe) to Amber (Warning). Push reminders are sent to the senior. Setting to 0 disables safety monitoring. |
| **Alert Timeout (Timer 2)** | 0 to 720 mins (Default: 60m grace, or 0) | **Mechanism:** The grace buffer after Timer 1 expires. **Consequence:** When Timer 2 expires, status flips to Red (Emergency). Push notifications and audible alarms are sent to all caregivers. Setting to 0 means emergency alerts fire immediately after Timer 1 expires. |
| **Vacation & Pause** | Date Range Picker (Start & | **Mechanism:** Suppresses timer execution during holidays or hospital stays. |

| Setting Name | Options & Ranges | Detailed Mechanism & Functional Consequence |
|---|---|---|
| **Monitoring** | End Date) | **Consequence:** All countdowns are paused. Caregiver dashboard shows a calm grey "Vacation" badge. Zero false alarms are dispatched during the selected window. |

## 4.2 Passive Activity & Sensor Automations

| Setting Name | Options & Ranges | Detailed Mechanism & Functional Consequence |
|---|---|---|
| **Battery Charger Activity** | Enabled / Disabled (Default: Enabled) | **Mechanism:** Listens for system battery plug-in / unplug broadcasts. **Consequence:** When elder plugs in phone at bedtime or unplugs in the morning, the system silently resets the inactivity timer! Elder is confirmed safe with zero manual app interaction. |
| **Screen Unlock Activity** | Enabled / Disabled (Default: Enabled) | **Mechanism:** Intercepts ACTION_USER_PRESENT lock-screen events. **Consequence:** If elder unlocks their phone to check photos, read news, or take a call, it registers as a valid "sign of life" and resets Timer 1. |
| **Apple Health / Google Fit Sync** | Enabled / Disabled (Default: Disabled) | **Mechanism:** Reads HealthKit / Google Fit step counts and active energy. **Consequence:** If the senior walks more than 100 steps in an hour or their Apple Watch records normal heart activity, it automatically logs activity. |
| **Fall Detection Sensitivity** | High (Carpet) Medium (Wood) Low (Tile) | **Mechanism:** Adjusts 3-axis accelerometer vector magnitude threshold. **Consequence:** "High" triggers on 1.8G impacts (soft carpet); "Medium" on 2.5G; "Low" on 3.2G. Always initiates a 30-second local cancel window before alerting caregivers. |
| **Shake to Panic Alert** | Enabled / Disabled (Default: Disabled) | **Mechanism:** Evaluates continuous high-frequency shake gestures. **Consequence:** Violently shaking the phone for 3 seconds triggers a 10-second panic countdown, then dispatches immediate emergency alerts to family contacts. |

## 4.3 Geofencing & Safe Zones

| Setting Name | Options & Ranges | Detailed Mechanism & Functional Consequence |
|---|---|---|
| **Enable Safe Zones** | Enabled / Disabled | **Mechanism:** Background GPS circular boundary monitoring. **Consequence:** When enabled, safety check-in countdowns are paused while inside home coordinates. If elder wanders outside the safe perimeter, monitoring activates or alerts caregivers. |

| Setting Name | Options & Ranges | Detailed Mechanism & Functional Consequence |
|---|---|---|
| **Safe Zone Coordinates** | Interactive Map Picker | **Consequence:** Pins the center of the safe zone (e.g. Senior's Residence). |
| **Safe Zone Radius** | 50m, 100m, 200m, 500m | **Consequence:** Sets the geographical radius. A 200m radius covers the elder's home and immediate garden/driveway. |

## 4.4 Display, Language & Caregiver Customization

| Setting Name | Options & Ranges | Detailed Mechanism & Functional Consequence |
|---|---|---|
| **App UI Language** | English / Traditional Chinese / Simplified Chinese | **Consequence:** Changes all navigation menus, dialogs, and button labels. Decoupled from Scripture language. |
| **Bible Translation** | Chinese CUV (和合本)<br>English NIV<br>English KJV | **Consequence:** Selects the offline scripture translation. Seniors can read Chinese scriptures while adult children use English UI. |
| **Auto-Minimize on Check-In** | Enabled / Disabled (Default: Enabled) | **Consequence:** If enabled, app auto-exits to home screen 500ms after tapping check-in. If disabled, app stays open on Senior Dashboard. |
| **Caregiver Alert Banner Flashing** | Enabled / Disabled (Default: Enabled) | **Consequence:** If enabled, Red alerts pulse rhythmically to catch attention. If disabled, banner remains solid. |
| **Live Activities (iOS Dynamic Island)** | Normal / Expired toggles & color picker | **Consequence:** Controls whether the Dynamic Island shows the countdown during normal hours or only when expired. |

CHAPTER 5 · CAREGIVER OPERATIONS & PROTOCOLS

# The Caregiver Console, Instant QR Pairing & Emergency Protocols

> How family members connect, monitor multiple loved ones, and handle clinical alert scenarios.

## 5.1 Instant AES QR Code Pairing (3-Second Setup)

1. **Senior Phone:** Open Settings → tap **"My Caregivers" (我的照護者)**. An encrypted, rotating QR code appears on screen.
2. **Caregiver Phone:** Open Caregiver Dashboard → tap the round blue **"+"** button → tap **"Scan QR Code"**.
3. **Scan:** Align the camera with the senior's QR code. Pairing completes in 3 seconds! Multiple siblings can scan the same code to form a care circle.

## 5.2 The Caregiver Dashboard & Multi-Senior Management

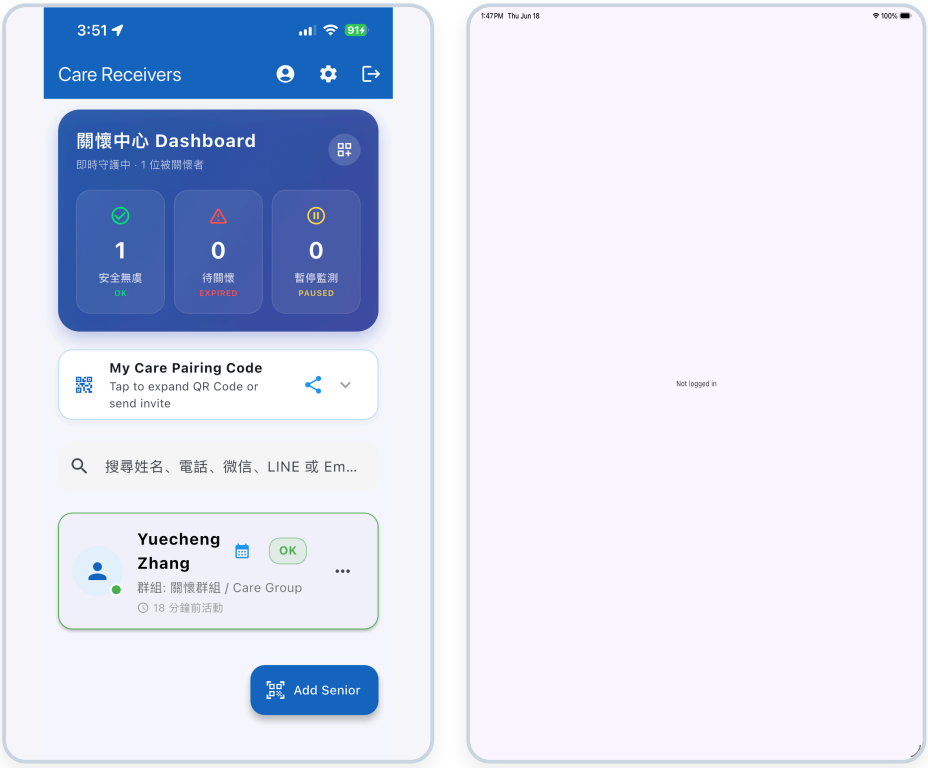



Figure 5.1: Caregiver Console on Phone (Left) and Multi-Senior iPad Grid (Right)

The Caregiver Console displays cards for each connected senior with real-time status badges, countdown clocks, last active timestamps, and direct phone/SMS shortcuts.

## 5.3 The Five Real-Time Conditions & Action Protocol

**Condition 1: GREEN · Safe & Checked-In**

**What it means:** The elder checked in within the active safe window (e.g. today at 08:30 AM via Apple Watch).

**Caregiver Action:** Total peace of mind! No intervention required. You can tap the "Send Warm Message" shortcut to share a greeting.

**Condition 2: AMBER · Warning / Overdue Buffer**

**What it means:** Timer 1 has expired, but Timer 2 grace buffer is counting down. The elder may be sleeping in or out gardening.

**Caregiver Action: Do not panic.** Make a casual phone call or send a gentle text: *"Good morning Mom, hope you have a wonderful day!"*

**Condition 3: RED · Emergency Safety Timeout**

**What it means:** Both Timer 1 and Timer 2 have expired with zero check-in signals. The app sounds an alarm and displays a persistent banner.

**Caregiver Action:**

1. Immediately tap the phone icon on the card to call the elder.
2. If unanswered, call a neighbor, building manager, or local relative.
3. If no contact can be made, dispatch an in-person safety check.

**Condition 4: RED · Automatic Fall Detected**

**What it means:** The elder's phone detected a heavy impact and the local 30-second cancellation window was not dismissed.

**Caregiver Action:** Call the elder immediately. If unanswered, call local emergency services (911 / 119) and head to the residence.

**Condition 5: GREY · Vacation / Monitoring Paused**

**What it means:** Vacation Mode is active for the scheduled dates.

**Caregiver Action:** All alarms are paused. Enjoy the family holiday or recovery period without false alarms.

CHAPTER 6 · DEVOTIONALS, BOUNDARIES & ROADMAP

# Daily Scriptures, Current Technical Boundaries & Future Roadmap

> Spiritual wellness integration, transparent engineering limits, and our 24-month roadmap.

## 6.1 Offline Bible Reader & Spiritual Health




Figure 6.1: Daily Scripture Card (Left) and Full Offline Chapter Reader (Right)

Elders start their morning with a calm devotional verse. Tapping the verse opens the 100% offline chapter reader supporting Chinese Union Version (和合本), English NIV, and English KJV with adjustable font sizes.

## 6.2 Current Implementation Boundaries

- **Internet Connectivity for Remote Sync:** Local check-ins, fall detection, nightstand clocks, and scriptures work completely offline; however, updating the remote caregiver dashboard requires Wi-Fi or cellular data.
- **Smart Speaker Hardware:** Voice check-in requires an Amazon Echo or Alexa app.
- **iOS Background Scheduling:** Continuous background accelerometer monitoring on iOS without an active HealthKit workout session is governed by Apple's battery throttling.

## 6.3 Eventual Product Roadmap (12 to 24 Months)

1. **Cellular Automated PSTN Voice & SMS Fallback:** If caregiver push notifications fail or internet is unavailable, cloud servers automatically dial real phone calls.
2. **Matter / 60GHz Millimeter-Wave Radar Sensors:** Camera-free ambient sensors detect presence and respiration without touching any device.
3. **AI Circadian Anomaly Detection:** Machine learning identifies deviations from natural routines (e.g. normally awake by 7:30 AM, inactive at 9:00 AM).
4. **Apple Watch Standalone Cellular Sync:** Direct Apple Watch cloud sync without requiring an iPhone nearby.